Cavity Blockade in the Jaynes-Cummings Model

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February 16, 2026

The Driven Jaynes-Cummings Model

The system consists of a two-level atom coupled to a cavity mode, driven by an external laser field $\mathcal{E}$.

Rotating Frame Hamiltonian (RWA):

$$H' = \delta_c a^\dagger a + \delta_a |e\rangle\langle e| + g(\sigma^+ a + \sigma^- a^\dagger) + \mathcal{E}(a + a^\dagger)$$

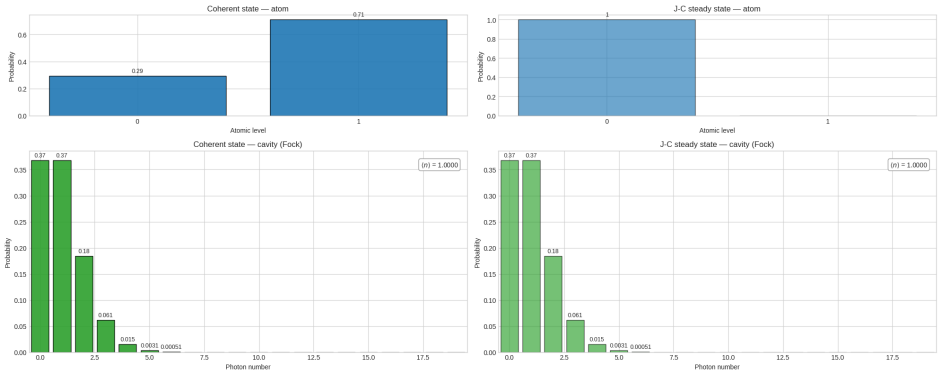
Physical Meaning of H_d:

- Represents a **coherent driving field** pumping the cavity.
- Realized experimentally by a laser beam of frequency ω.
- $\mathcal{E}$ characterizes the laser intensity/amplitude.

Linear Regime: The Coherent State ($g = 0$)

In the absence of atom-cavity coupling, the system behaves as a driven damped harmonic oscillator.

- Steady state is a **coherent state** $|\alpha\rangle$.
- Occupation follows a **Poisson distribution**.

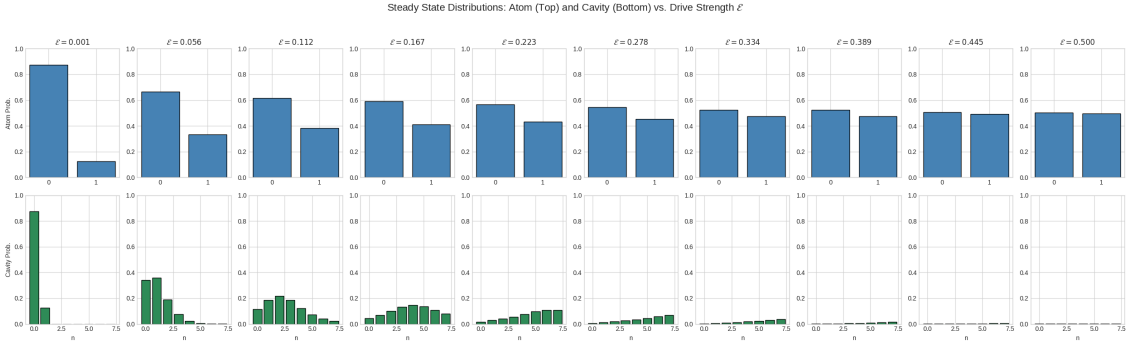


Result: Distribution matches the analytical Poissonian curve.

The Photon Blockade Effect

In the strong coupling regime ($g \gg \kappa, \Gamma$), the J-C ladder becomes **anharmonic**.

- **Mechanism:** Tuning ω to the first polariton resonance ($\omega_c \pm g$) allows one photon to enter the cavity.
- **The Block:** The transition to the 2-photon state is detuned by $(\sqrt{2}-1)g$, preventing multi-photon absorption.



Observation: $P(n \geq 2)$ remains negligible even as drive $\mathcal{E}$ increases.

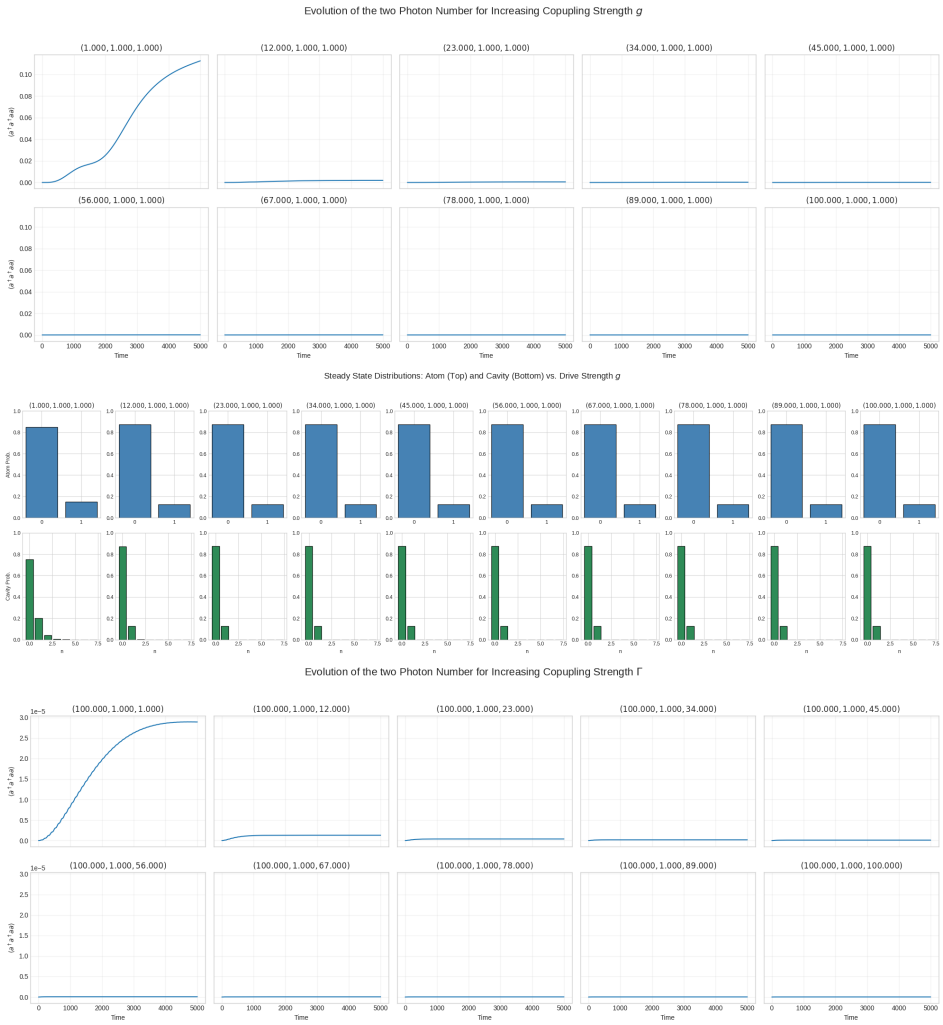
Sensitivity to Dissipation and Coupling

Coupling g:

- Larger g increases anharmonicity and improves blockade quality.

Dissipation κ, Γ:

- Increasing decay rates broadens peaks.
- Is harder to keep photons in the cavity, reducing blockade effectiveness.



Analysis: Low κ is essential for resolved quantum states.

Quantifying the Blockade: $g^{(2)}(0)$

The second-order correlation function measures the probability of detecting two simultaneous photons.

$$g^{(2)}(0) = \frac{\langle a^\dagger a^\dagger a a \rangle}{\langle a^\dagger a \rangle^2}$$

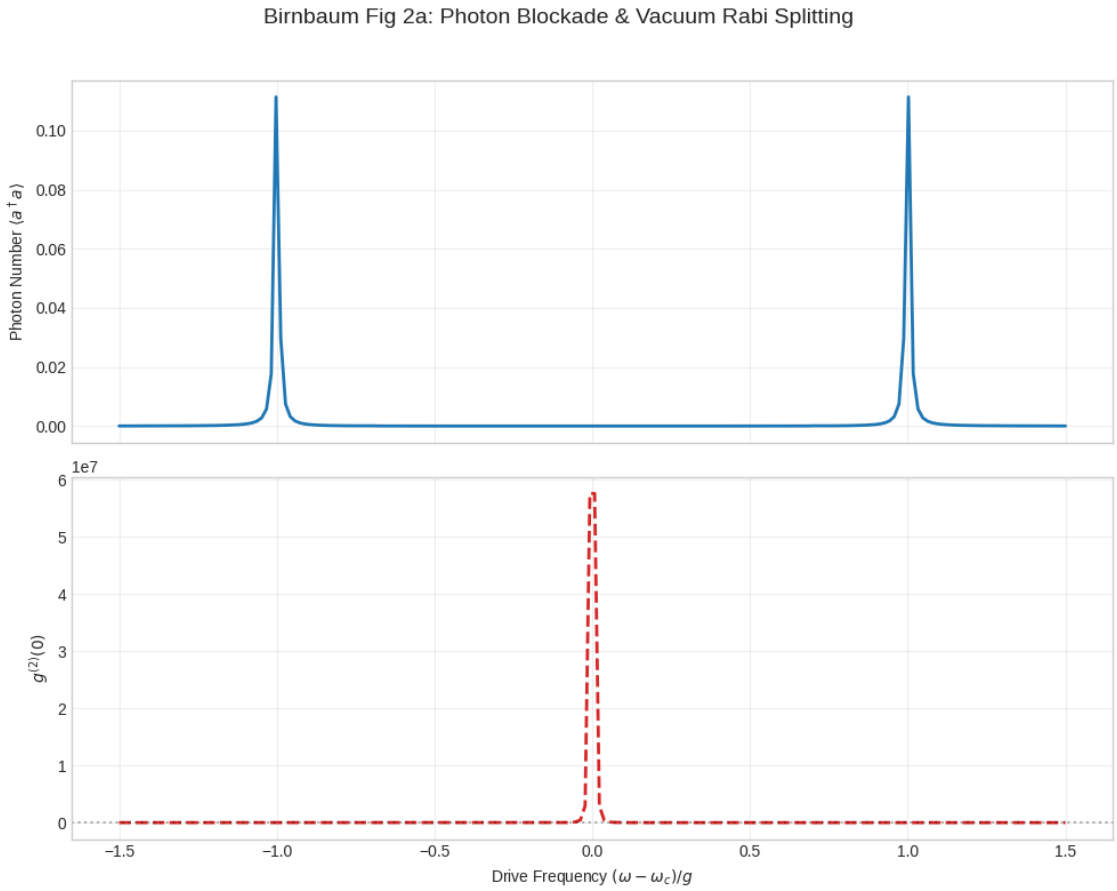
Numerical Results ($g = 100$):

- $\langle n \rangle \approx 0.125$
- $g^{(2)}(0) \approx 0.0018$

Conclusion: $g^{(2)}(0) \ll 1$ confirms it. The field is a source of single photons.

Spectroscopy: Vacuum Rabi Splitting

Scanning the drive frequency ω reveals the dressed states of the system.



- Matches experimental results (e.g., Birnbaum et al.).